Listing 10-4. *Serial Time Clock Setting Source Code*

```
#include <Wire.h>

const int DS1307 = 0x68;

const char* days[] =
  {"Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday"};
const char* months[] =
  {"January", "February", "March", "April", "May", "June", "July", "August",
   "September", "October", "November", "December"};

byte second = 0;
byte minute = 0;
byte hour = 0;
byte weekday = 0;
byte monthday = 0;
byte month = 0;
byte year = 0;

void setup() {
  Wire.begin();
  Serial.begin(9600);

  delay(2000);
  Serial.print("The current date and time is: ");
  printTime();
  Serial.println("To set the date and time please select Newline ending to continue.");
  Serial.println("Would you like to set the date and time now? Y/N");
  while (!Serial.available()) delay(10);
  if (Serial.read() == 'y' || Serial.read() == 'Y') {
    Serial.read();
    setTime();
    Serial.print("The current date and time is now: ");
    printTime();
  }
  Serial.println("Goodbye.");
}

void loop() {}

byte decToBcd(byte val) {
  return ((val/10*16) + (val%10));
}

byte bcdToDec(byte val) {
  return ((val/16*10) + (val%16));
}
```